Shailza Jolly

shailzajolly@gmail.com | 0176 67279750 | Berlin, Germany | shailzajolly | shailzajolly

Summary

- ML Scientist / AI Engineer with a Ph.D. in Computer Science and 6+ years across industry research and product, focused on making LLM systems accurate and auditable enough to trust. Led a data-quality pipeline over billions of tokens of LLM training data at Amazon AI; built LLM-powered product features for a platform serving medical device manufacturers at Flinn.ai; open-sourced Sift, a hybrid BM25 and dense retrieval system whose agent answers only from retrieved passages, with a source and page citation on every claim. Working depth in modern LLM internals and inference efficiency (KV-cache, speculative decoding, FlashAttention) and in agentic system design (MCP, persistent memory, tool orchestration). Publications at AAAI, NAACL, and EMNLP.

Experience

Independent, Applied AI & LLM Systems — Berlin, Germany — July 2025 – present — 1 year 1 month

- Built and open-sourced Sift: grounded question-answering over technical PDF corpora. Docling chunking, hybrid BM25 and dense (Chroma) retrieval, cross-encoder reranking, served over MCP to an agent that answers only from retrieved passages, with a source and page citation on every claim.
- Built an agentic tutoring system in daily use: an LLM agent orchestrating retrieval over a structured knowledge base, persistent memory of errors and progress, adaptive lesson selection that targets weak areas, and voice I/O (faster-whisper ASR, edge-tts TTS).
- Published technical writing on LLM internals and systems: attention mechanisms, data quality for LLM training, and the shift from generative to agentic AI.
- Deepened modern LLM architecture and inference-efficiency internals through Stanford coursework: KV-cache, speculative decoding, FlashAttention.

Flinn.ai, Senior AI Engineer — Berlin, Germany — Jan 2025 – June 2025 — 6 months

- Led development of LLM-powered product features for a platform serving medical device manufacturers, including data extraction from medical research papers and multilingual complaint monitoring.
- Partnered with product and backend teams to scope problems, define success metrics, and deliver features end to end.
- Navigated cost-performance tradeoffs under production constraints.

Parental Leave, Career break — Berlin, Germany — Dec 2023 – Dec 2024 — 1 year 1 month

Amazon AI, Research Scientist — Berlin, Germany — July 2022 – Nov 2023 — 1 year 5 months

- Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
- Presented findings and recommendations to senior science and engineering leadership; mentored Master's and Ph.D. interns on research deliverables.

German Research Center for Artificial Intelligence (DFKI), Machine Learning Scientist — Berlin, Germany — Mar 2019 – June 2022 — 3 years 4 months

- Implemented transformer-based ML systems end-to-end (GPU training, Docker, production-level code) within BMBF-funded projects; published results at AAAI, NAACL, and EMNLP.
- Collaborated with academic and industry partners internationally; supervised interns and BSc/MSc students.

NVIDIA Research, Machine Learning Scientist Intern — Santa Clara, US — May 2021 – Aug 2021 — 4 months

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa, Applied Scientist Intern — Aachen, Germany — Aug 2019 – Dec 2019 — 5 months

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

SAP AI Research, Research Intern / Master's Thesis — Berlin, Germany — May 2018 – Feb 2019 — 10 months

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

Skills

ML/GenAI: LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation, model training and fine-tuning (T5, BERT)

LLM Systems: Agentic workflows (MCP, LangGraph/LangSmith), Retrieval-Augmented Generation (RAG), grounding and citations, persistent agent memory, experiment design, offline evaluation

Search & Retrieval: Hybrid retrieval (BM25 + dense), cross-encoder reranking, vector search (Chroma, Pinecone), Sentence Transformers, Docling

LLM Internals & Efficiency: KV-cache, speculative decoding, FlashAttention

Frameworks & Libraries: Python, PyTorch, Hugging Face (Transformers, Datasets), PySpark, Weights & Biases

Infrastructure & Deployment: AWS, Docker, FastAPI, Git

Education

Ph.D. **TU Kaiserslautern**, Computer Science — Kaiserslautern, Germany — 2019 – 2022

- Grade: Sehr Gut 1.0 (highest on 1.0–5.0 scale)
- Thesis: Building Natural Language Generation and Understanding Systems in Data-Constrained Settings

Visiting Researcher **University of Copenhagen**, Natural Language Processing — Copenhagen, Denmark — Jan 2021 – Mar 2021

- Developed an unsupervised post-editing algorithm to generate fluent fact-checking explanations

M.Sc. **TU Kaiserslautern**, Computer Science — Minor in Economics — Kaiserslautern, Germany — 2016 – 2018

- Grade: Sehr Gut 1.5

Semester Abroad **Kyushu University**, Computer Science — Fukuoka, Japan — Nov 2017 – Feb 2018

- Explainable AI to analyze behavior of deep CNN architectures for image recognition

B.Tech **Guru Nanak Dev Engineering College**, Computer Science & Engineering — India — 2012 – 2016

- Grade: First Division with Distinction

Selected Publications

Search and Learn: Improving Semantic Coverage for Data-to-Text Generation — Jan 2022
Shailza Jolly, Z. X. Zhang, A. Dengel, L. Mou
arxiv.org/abs/2112.02770 (AAAI)

EaSe: A Diagnostic Tool for VQA Based on Answer Diversity — Jan 2021
Shailza Jolly, S. Pezzelle, M. Nabi
www.aclweb.org/anthology/2021.naacl-main.192 (NAACL-HLT)

Can Pre-training Help VQA with Lexical Variations? Jan 2020
Shailza Jolly, S. Kapoor
www.aclweb.org/anthology/2020.findings-emnlp.257 (Findings of EMNLP)

Data-Efficient Paraphrase Generation to Bootstrap Intent Classification and Slot Labeling for New Features in Task-Oriented Dialog Systems Jan 2020
Shailza Jolly, T. Falke, C. Tirkaz, D. Sorokin
www.aclweb.org/anthology/2020.coling-industry.2 (COLING (Industry Track))

How Do Convolutional Neural Networks Learn Design? Jan 2018
Shailza Jolly, B. K. Iwana, R. Kuroki, S. Uchida
ieeexplore.ieee.org/abstract/document/8545624 (International Conference on Pattern Recognition (ICPR))

Awards

AAAI-22 Scholarship by Hitachi Jan 2022

AI Newcomer 2021 Award Jan 2021
- Recognized by the German Informatics Society and BMBF, Germany.

EU-Cost STSM Grant Jan 2020
- Awarded by EU-Cost Action to work on the Multi3generation project at CopeNLU, University of Copenhagen.

Best Student Paper Award Jan 2018
- Awarded by the International Conference on Pattern Recognition (ICPR) for "How Do Convolutional Neural Networks Learn Design?"

Media Coverage

- Interview featured in the Rheinfalz newspaper.
- Radio interview for Antenne Kaiserslautern, Germany.

Languages

English: Fluent
German: A2 (actively learning)
Hindi: Native speaker